

Last week we determined the resistance of three resistors by measuring the potential and current of each resistor and then using Ohm's Law with the data. This week we will connect two of these resistors in series and in parallel, measure the potential and current for the pair together to determine the **total resistance** for the configuration.

Ohm's Law states that the electric current flowing through a *resistor* is proportional to the *potential difference* across the resistor and inversely proportional to the *resistance* of the resistor. This can be written as a simple equation, where I is the electric current, V is the potential difference, and R is the resistance:

$$I = \frac{V}{R} \quad \text{or} \quad V = I R$$

For two resistors in series, we expect that the total resistance for the pair together is equal to the sum of the individual resistances. That is:

$$\text{resistors in series: } R_{total} = R_A + R_B$$

For two resistors in parallel, we expect that the total resistance for the pair together is equal to the product of the individual resistances divided by the sum of the individual resistances. That is:

$$\text{resistors in parallel: } R_{total} = \frac{R_A R_B}{R_A + R_B}$$

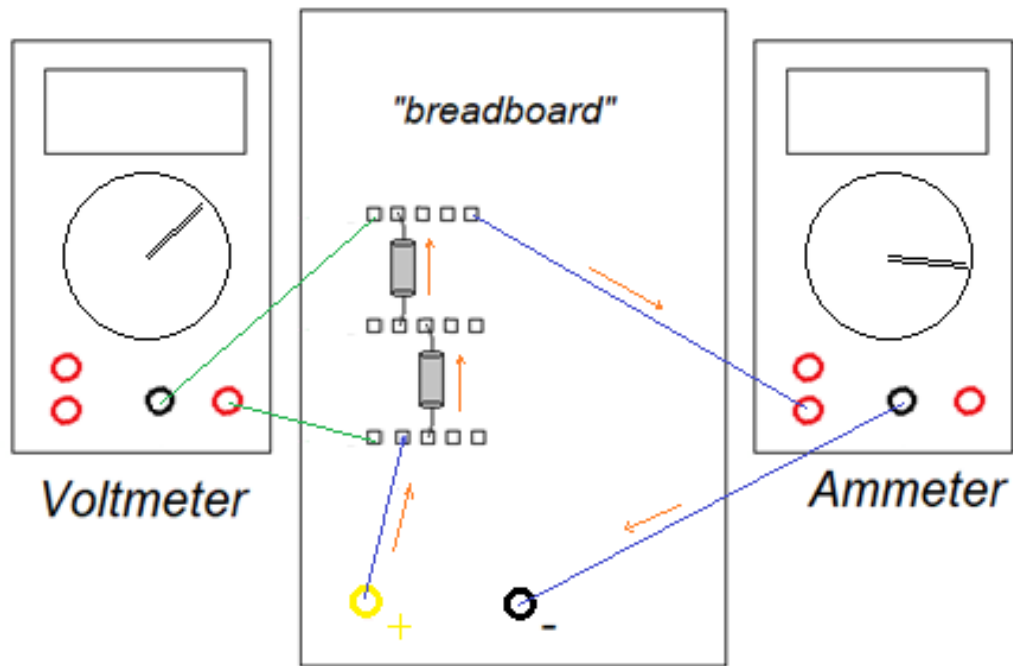
Part 2: Resistors in Series or Parallel

The procedure to collect data for this part is identical to that of Part 1. The only difference is that we will use a pair of resistors for each trial instead of an individual resistor.

Perform this procedure for **resistors in series** and then for **resistors in parallel**:

- For each of three **pairs of resistors**, record ten values of the potential and corresponding current.
- Create a data table of current in *mA* and potential in *Volts* for each of the three pairs of resistors.
- Create one graph, V vs. I , for your three sets of data (just as you did in Lab 5).
- The slope of the best fit line to each set of data is the *measured* value of the total resistance of the corresponding pair of resistors. Calculate the expected resistance for each pair and compare each measured to its corresponding expected.

Resistors in Series



Resistors in Parallel

